

Junze Zhu

Ph.D. Candidate | Artificial Intelligence, Nanjing University

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Research interests

Language agents, multi-agent systems, speculative and asynchronous execution, harness-aligned policy learning, knowledge graphs, and recommendation.

Education

2025–2029 (expected) **Ph.D. Candidate in Artificial Intelligence**

Nanjing University
Focus: Agent Systems and Swarm Intelligence

2022–2025 **Master's in Data Science**

Wuhan University
Focus: Knowledge-Enhanced Intelligent Systems

2018–2025 **Bachelor's in Electronic Information**

Wuhan University
Focus: Multimodal Fusion Perception

Selected publications

2026 Recognize Your Orchestrator: An Entropy Dynamics Perspective for LLM Multi-Agent Systems
Junze Zhu, Weihao Chen, Xuanwang Zhang, Zhen Wu, Xinyu Dai. *ICML 2026*, PMLR 306:166844–166869. Paper.

Mean-field entropy dynamics for understanding instability in centralized language-agent orchestration.

2025 Knowledge Graph Preference Contrastive Learning for Recommendation

Junze Zhu, Zhongyi Hu, Fan Zhang. *Learning on Graphs Conference*, PMLR 269:18:1–18:15. Paper / Code.
Preference-aware contrastive learning and rationale attention for knowledge-aware recommendation.

ICLR 2027 submissions

- **WebProgramBench: Can Coding Agents Reconstruct Live Applications from Scratch via Black-Box Interaction?** Xuanwang Zhang, Yuteng Han, Xinyu Liu, **Junze Zhu**, et al.; fourth author. *Under review at ICLR 2027*. The benchmark evaluates whether coding agents can reconstruct executable applications from black-box interaction, with validation across interfaces, services, and persistent state. OpenReview.
- **Perseus: Non-Blocking Parallel Exploration with a Speculative Swarm.** *Under review at ICLR 2027*. An asynchronous agent swarm architecture that broadens inference-time exploration through independent speculative branches while an authoritative Actor retains task control; ready-only evidence admission keeps the main trajectory coherent. Project site.
- **Harness Policy Distillation: Aligning Agent Policies with Runtime Semantics.** *Under review at ICLR 2027*. A policy learning framework that distills verified actions and runtime control-document views into a harness-compatible policy, connecting model behavior with the semantics of the environment in which it executes.

Selected systems work

- **Open Crowd** – skill and CLI management for complex task orchestration with agent swarms.
- **Token-Never-Sleeps** – long-horizon agent swarm task management and external-loop execution; published as an npm package.

Links

Academic homepage | GitHub | Google Scholar | WebProgramBench on OpenReview